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**Comparison of Eddy Viscosity -
Transport Turbulence Models for
Three-Dimensional, Shock-Separated
Flowfields**

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COMPARISON OF EDDY VISCOSITY - TRANSPORT TURBULENCE MODELS FOR THREE-DIMENSIONAL, SHOCK-SEPARATED FLOWFIELDS

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1. Abstract

An evaluation of several one-equation "eddy viscosity transport" turbulence closure models as applied to three-dimensional shock wave / boundary layer interactions is presented herein. Comparisons among a version of the Baldwin-Barth model, an approach of Edwards and McRae, a nonequilibrium model of Edwards and Chandra, and a modified form of the Spalart-Allmaras model are presented for two test cases, one involving Mach 8 flow over a flat plate / sharp fin apparatus and one involving Mach 3 flow over a cylinder - offset flare geometry. Strengths and weaknesses of the one-equation approaches are highlighted through direct comparison with experimental data, and the effects of grid refinement are examined.

2. Introduction

Several works¹⁻⁶ have advocated the "eddy viscosity - transport" approach as a means of closing the Reynolds-averaged Navier-Stokes system for complex, wall-bounded flows. The original concept of a single transport equation governing directly the development of the eddy viscosity field was proposed by Nee and Kovasznay¹ in the late 1960's. After lying dormant in this country for twenty years, the idea was rediscovered by Baldwin and Barth², who developed a near-wall calibration procedure and an efficient strategy for advancing the solution of the eddy viscosity equation. Since the publication of Ref. 2, there has been steady progress in the development of eddy viscosity - transport methods in the United States. Spalart and Allmaras at Boeing³, Durbin and Yang at CTR⁴, and the present author at NCSU / NC A&T SU^{5,6} have all derived variations on the basic theme, each with its own degree of complexity and applicability. Independent testing of some of these approaches has been performed by Menter⁷ for low Mach number adverse pressure gradient flows, by Kandula and Buning⁸ for exhaust jet flowfields, by Birch⁹ for mixing layers, and

by Goldberg¹⁰ for a wide variety of flows. None of the flows considered in the tests were three-dimensional, and, except for a few computations by Goldberg, all were at either subsonic or transonic conditions.

Basic comparisons among several of the approaches for 2-D supersonic / hypersonic shock-boundary layer interactions have been presented in Ref. 6. Several general deficiencies of the various eddy viscosity - transport methodologies were delineated, with the most notable being a failure to predict the proper recovery of a turbulent boundary layer following a strong disturbance. It was noted, however, that the baseline models (Baldwin-Barth, Spalart-Allmaras, Edwards-McRae) were generally able to predict both the upstream extent of axial separation and the wake-like velocity distribution above the backflow region quite well. In an attempt to rectify the reattachment / recovery problem, a nonequilibrium version of the Edwards-McRae model (referred to as the Edwards-Chandra model in this work) was derived and tested. The Edwards-Chandra approach differs from the Edwards-McRae formulation in its inclusion of variable closure coefficients, dependent on several measures of the departure of the wall-bounded flow from local equilibrium.

In this paper, we attempt to ascertain the degree of accuracy and predictive capability offered by the four eddy viscosity - transport approaches for three-dimensional flow situations involving shock wave - turbulent boundary layer interaction. Two recently-obtained experimental databases are considered as a means toward achieving this goal. The first database corresponds to an experiment by Kussoy and Horstman¹¹ and involves Mach 8 flow over a flat plate / sharp fin geometry. The second database corresponds to an experiment by Wideman, et al.¹² and involves Mach 3 flow over a cylinder - offset cone configuration. Comparisons with the experimental data, both in regard to surface measurements and to the postulated flowfield structure, will be used to delineate the relative strengths and weaknesses of the closure concepts for these flows. The effects of grid refinement on the obtained solutions and convergence rates will also be examined.

3. Governing Equations, Discretization Approach, and Numerical Method

The flowfields under consideration are assumed to

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be governed by the three-dimensional, compressible, Reynolds-averaged Navier-Stokes equations. In a curvilinear coordinate system defined by the steady transformation $\xi = \xi(x, y, z)$, $\eta = \eta(x, y, z)$, $\zeta = \zeta(x, y, z)$, the three-dimensional Navier-Stokes equations can be written in strong conservation law form as

$$\frac{\partial \hat{U}}{\partial t} + \frac{\partial(\hat{E} - \hat{E}_v)}{\partial \xi} + \frac{\partial(\hat{F} - \hat{F}_v)}{\partial \eta} + \frac{\partial(\hat{G} - \hat{G}_v)}{\partial \zeta} = 0 \quad (1)$$

where $\hat{U}$ is the vector of conserved variables, $\hat{E}$, $\hat{F}$, and $\hat{G}$ are the inviscid fluxes, and $\hat{E}_v$, $\hat{F}_v$ and $\hat{G}_v$ are the viscous fluxes. Reynolds stress effects are incorporated by invoking the Boussinesq hypothesis, thus reducing the problem to the specification of an eddy viscosity and a turbulent Prandtl number (0.9 in this work). Compressibility effects are assumed to result from variations in the mean density.

A control volume discretization of the Navier-Stokes set is adopted, with a second-order MUSCL extension of the Wada-Liou upwinding scheme¹³ used for the inviscid fluxes. The Wada-Liou approach is a hybrid flux-vector / flux-difference splitting technique that is capable of capturing shear layers without excess numerical diffusion. The viscous components of the Navier-Stokes set are central-differenced to second-order accuracy.

The 3-D Reynolds-averaged Navier-Stokes set is solved using a recently-developed upwind relaxation multigrid algorithm.¹⁴ This approach, an extension of the nonlinear relaxation method described in Ref. 5, is capable of providing rapid convergence for three-dimensional, shock-separated flowfields.

4. Turbulence Model Descriptions

In this section, the Baldwin-Barth model, the Edwards-McRae approach, and the Edwards-Chandra modification are presented in a common notation. A modified variant of the Spalart-Allmaras model, incorporating an alternative calibration procedure for the viscous sublayer, is also described. For comparative purposes, the notation for the latter development follows that of Ref. 3.

4.1 Basic descriptions

In terms of the transported quantity $F \equiv k^2/\epsilon$, the eddy viscosity field for the Baldwin-Barth, Edwards-McRae, and Edwards-Chandra models is defined as

$$\mu_t = \rho C_\mu F D_1 D_2 \equiv \rho \nu_t \quad (2)$$

where C_μ is a constant and D_1 and D_2 are wall-damping functions of Van Driest form.

$$D_i = 1 - \exp\left(-\frac{\hat{n}^+}{C_i^+}\right) \quad (3)$$

The distribution of F is determined by the solution of

$$B_1 \frac{DF}{Dt} = B_2 f_2(D_1, D_2) \sqrt{F \nu_t} S + \left(\nu + \frac{\nu_t}{\sigma_R}\right) \nabla^2 F - \frac{1}{\sigma_\epsilon} (\nabla \nu_t) \cdot \nabla F \quad (4)$$

for the Baldwin-Barth model and

$$B_1 G_a \frac{DF}{Dt} = B_2 G_p f_2(D_1, D_2) \sqrt{F \nu_t} S + \frac{1}{\sigma_R} (\nu + \nu_t) \nabla^2 F - \frac{1}{\sigma_\epsilon} \frac{\nu_t F}{\hat{n}^2} \quad (5)$$

for the Edwards-McRae / Edwards-Chandra approaches. In the preceding expressions, B_1 , B_2 , σ_R , and σ_ϵ are constants, ν is the kinematic molecular viscosity, $\hat{n}$ is a representative distance from the wall (or walls), and

$$S = \left(\frac{\partial \vec{U}_i}{\partial x_j} + \frac{\partial \vec{U}_j}{\partial x_i}\right) \frac{\partial \vec{U}_i}{\partial x_j} - \frac{2}{3} \left(\frac{\partial \vec{U}_k}{\partial x_k}\right)^2$$

The wall damping function $f_2(D_1, D_2)$ is given as

$$f_2 = \left(1 + \frac{\mu}{\mu_t}\right) (D_1 D_2)^{\frac{3}{2}} \left[1 + \hat{n}^+ \left(\frac{\exp(-\hat{n}^+/C_1)}{C_1 D_2} + \frac{\exp(-\hat{n}^+/C_2)}{C_2 D_1}\right)\right] \quad (6)$$

for the Baldwin-Barth model and as

$$f_2 = \left(1 + \frac{\mu}{\mu_t}\right) (D_1 D_2)^{\frac{3}{2}} \quad (7)$$

for the Edwards-McRae and Edwards-Chandra models. The wall coordinate $\hat{n}^+$ is defined as

$$\hat{n}^+ = \frac{1}{\nu} \left(\frac{C_\mu F}{\kappa} + \beta U \hat{n}\right) \quad (8)$$

where β is a constant and U is the magnitude of the flow velocity vector $\vec{U}$. In the log region and below, $F = \frac{\kappa U_\tau \hat{n}}{C_\mu}$, and $\hat{n}^+$ reduces to its more conventional form $\frac{U_\tau \hat{n}}{\nu}$. The second term in Eq. 8 allows some control over the "numerical" transition length provided by the models.⁶

The terms in (4) and (5) represent, respectively, convection, production, diffusion, and destruction of the quantity F . It is obvious that the major differences between (4) and (5) lie in the modeling of the destruction term. In common with the Nee-Kovasznay and Spalart-Allmaras formulations, the Edwards-Chandra expression maintains a direct dependence on the normal distance from the wall (or walls) $\hat{n}$. Values for the modeling constants in Eqns. 2-8 are given as follows: (bracketed quantities refer to the Baldwin-Barth formulation)

$$\begin{aligned} \frac{1}{\sigma_\epsilon} &= \frac{B_2 \sqrt{C_\mu}}{\kappa^2}, & \frac{1}{\sigma_R} &= 1.64 \frac{1}{\sigma_\epsilon} [1.00 \frac{1}{\sigma_\epsilon}], & B_1 &= 1.0[0.4], \\ B_2 &= 1.0[0.8], & C_1^+ &= 19[26], & C_2^+ &= 10, \\ C_\mu &= 0.09, & \kappa &= 0.41 & \beta &= .005 \end{aligned} \quad (9)$$

Further details regarding the determination of the wall-damping functions and the calibration procedures can be found in Refs. 2 and 6.

4.2 Edwards-Chandra modifications

For the Edwards-McRae formulation, the functions G_a and G_p in Eq. 5 are equal to unity. For the Edwards-Chandra modification, G_a and G_p vary in response to departures of the flow from local equilibrium, thereby adjusting the relative balances of the terms within the transport equation. Such departures from equilibrium are measured by considering the behavior of limited forms of velocity scale ratios obtainable within the one-equation framework:

$$F1 = \max(1.0, \frac{\sqrt{(\nu + \nu_t)S^{1/2}}}{U_\tau}) - 1.0, \quad (10)$$

$$F2 = \max(1.0, \frac{C_\mu F}{\kappa U_\tau \hat{n}}) - 1.0 \quad (11)$$

and

$$F3 = \tanh[\alpha_3(\frac{U}{\sqrt{(\nu + \nu_t)S^{1/2}D_1D_2}})^{k_3}], \quad (12)$$

In the above, the friction velocity U_τ is defined by

$$U_\tau = \max(\sqrt{\nu S^{1/2}}|_{wall}, \frac{C_\mu F_{wall+1}}{\kappa(\hat{n}_{wall+1} - \hat{n}_{wall})}) \quad (13)$$

where the notation $wall+1$ refers to the grid point immediately above the location of the solid wall. Within a particular profile of an equilibrium turbulent boundary layer, $F1$ and $F2$ are identically zero, and $F3$ is equal to unity. Drastic departures from this behavior occur for nonequilibrium situations induced by a strong adverse pressure gradient in the streamwise direction. In particular, $F1$ and $F2$ rise above their equilibrium values, while $F3$ decays within a region of axially-separated flow. Based on these trends, we multiply the production term of Eq. 5 by the following function of $F1$ and $F3$ to simulate the observed amplification of the Reynolds stresses in response to a strong disturbance:

$$G_p = 1.0 + \alpha_p F3 F1 \quad (14)$$

The growth of G_p is opposed in the separated zone by the decay of $F3$. To allow the nonequilibrium Edwards-Chandra model to predict the same level of upstream influence as the baseline Edwards-McRae approach ($G_a = G_p = 1.0$), it is necessary to multiply the advective term in Eq. 5 by a similar function:

$$G_a = 1.0 + \alpha_a (F2 + F1 - \tanh(F2) F1) \quad (15)$$

The modeling constants used in the expressions for G_p and G_a are given as follows:

$$\begin{aligned} \alpha_3 &= 0.02, & k_3 &= 3, \\ \alpha_p &= 1.3, & \alpha_a &= 2.5 \end{aligned} \quad (16)$$

Further discussion regarding the construction and calibration of the flow-dependent coefficients can be found in Ref. 6.

4.3 Spalart-Allmaras modifications

A modified version of the Spalart-Allmaras model is also considered in the comparative study. For this approach, the eddy viscosity field is defined as

$$\mu_t = \rho \tilde{\nu} f_{v1} \equiv \rho \nu_t \quad (17)$$

where the damping function f_{v1} is given by

$$f_{v1} = \frac{\lambda^3}{\lambda^3 + c_{v1}^3}, \quad (18)$$

$$\lambda \equiv \frac{\tilde{\nu}}{\nu} \quad (19)$$

The distribution of $\tilde{\nu}$ is determined by the solution of

$$\begin{aligned} \frac{D\tilde{\nu}}{Dt} &= c_{b1}\tilde{S}\tilde{\nu} + \frac{1}{\sigma}[\nabla \cdot ((\nu + \tilde{\nu})\nabla\tilde{\nu}) \\ &+ c_{b2}(\nabla\tilde{\nu})^2] - c_{w1}f_w(r)(\frac{\tilde{\nu}}{\hat{n}})^2 \end{aligned} \quad (20)$$

Differences between the modified Spalart-Allmaras model and the original version relate mainly to the modeled near-wall behavior of the strain-rate norm $\tilde{S}$. In the original version³,

$$\tilde{S} = S^{1/2} + \frac{\tilde{\nu}}{(\kappa\hat{n})^2} f_{v2}, \quad (21)$$

$$f_{v2} = 1 - \frac{\lambda}{1 + \lambda f_{v1}} \quad (22)$$

The argument of the “wall-blockage” function f_w is defined using $\tilde{S}$:

$$r = \frac{\tilde{\nu}}{(\kappa\hat{n})^2\tilde{S}} \quad (23)$$

This form for $\tilde{S}$ ensures that the strain-rate norm maintains its logarithmic value $\frac{U_\tau}{\kappa\hat{n}}$ all the way to the wall. In our original implementation of the Spalart-Allmaras model⁶, the combination of the singular behavior of $\tilde{S}$ in the near-wall region, the lack of an exact linearization of $\tilde{S}$ (and f_w) in the implicit formulation of (21), and the large time steps allowed by the upwind relaxation flow solver produced oscillatory, limit-cycle convergence behavior in the Navier-Stokes residual norm after an initial period of rapid decrease. Such behavior made it extremely difficult to judge the progress of the solution in any rigorous sense; flow properties such as skin friction and heat transfer had to be monitored instead.

A more stable way of accounting for the behavior of $\tilde{S}$ in the laminar sublayer can be derived by applying ideas from Refs. 2 and 6. The result is given by the following redefinitions:

$$\tilde{S} = S^{1/2}(\frac{1}{\lambda} + f_{v1}), \quad (24)$$

$$r = \tanh[\frac{\tilde{\nu}}{(\kappa\hat{n})^2\tilde{S}}]/[\tanh(1.0)] \quad (25)$$

The modified Spalart-Allmaras model allows smooth, rapid convergence while retaining the near-wall accuracy of the original formulation. Additional constants and functions used in both Spalart-Allmaras formulations are given as follows:

$$f_w(r) = g \left[\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6} \right]^{1/6}, \quad (26)$$

$$g = r + c_{w2}(r^6 - r) \quad (27)$$

$$\begin{aligned} c_{b1} &= 0.1355, & \sigma &= \frac{2}{3}, \\ c_{b2} &= 0.622, & c_{w1} &= \frac{c_{b1}}{\kappa^2} + \frac{(1+c_{b2})}{\sigma}, \\ c_{w2} &= 0.3, & c_{w3} &= 2, \\ c_{v1} &= 7.1 \end{aligned} \quad (28)$$

5. Implementation Details

Equations 4, 5, and 20 are first transformed to a computational plane and are then discretized in the manner suggested by Baldwin and Barth, with first-order upwind differences used for the advective terms and second-order central differences used for the diffusion terms. For the Baldwin-Barth model, the destruction term is treated as outlined in Ref. 2 to maintain the positivity of the linear discretization operator. The choice of the destruction term in the Spalart-Allmaras formulation automatically maintains positivity. In all cases, the solution is advanced in time via a modified Euler implicit formulation, with the production term treated in a completely explicit fashion. In the evaluation of $\hat{n}$, multiple wall effects are accounted for by first measuring the distances from each wall to the point in question and then determining the minimum distance. The Edwards-Chandra model also requires an evaluation of the friction velocity U_τ to ensure proper scaling of the functions (10) and (11). For multiple wall cases, the "correct" friction velocity at a particular grid point is determined by first evaluating Eq. 13 along each of the walls and then choosing the value corresponding to the minimum distance $\hat{n}$. Each turbulence equation is advanced in the finest grid using one sweep of a planar Gauss-Seidel technique after each completed cycle of the mean-flow multigrid iteration procedure.

6. Results

6.1 Kussoy-Horstman sharp fin

Two recently-obtained experimental databases involving three-dimensional shock - boundary layer interaction are used to evaluate the performance of the eddy viscosity - transport approaches. The first database corresponds to an experiment by Kussoy and Horstman¹¹ and involves the Mach 8.2 interaction of a developing flat-plate turbulent boundary layer with the planar shock surface generated by a 15° fin placed perpendicular to the plate. Pertinent free-stream and inflow boundary layer properties for the interaction are summarized in the following table. The apex of the fin ($X = 0$ in the present

coordinate system) is located 176 cm downstream of the flat plate leading edge, and the boundary layer was surveyed (in the absence of the fin) at $X = 11$ cm.

	Kussoy-Horstman 15° fin
Mach no.	8.20
Re/meter	5.32×10^6
T_∞ (°K)	76.89
T_{wall}/T_∞	3.90
Inflow δ (cm)	3.7
Inflow θ (cm)	9.4×10^{-2}
Inflow C_f	1.0×10^{-3}
Inflow q_w (W/cm ²)	1.04

The inflow conditions for each computation were determined by integrating the 2-D Reynolds-averaged Navier-Stokes equations over a flat plate until the experimental momentum thickness θ was matched to sufficient accuracy.

The baseline grid used for the 3-D computations consists of 97 points in the streamwise (X) direction, 65 points normal to the flat plate (Y), and 89 points normal to the fin surface (Z). A finer grid consisting of 113x97x113 nodes is also used to assess the grid-dependence of the obtained solutions and convergence rates. The degree of clustering in the Y and Z directions is such that a minimum $\hat{n}^+$ of 1.0 or smaller is maintained in the equilibrium regions upstream of the interaction. The grid resolution utilized in the present investigation represents a significant improvement over that of Knight, et al.¹⁵ (64x40x64) in an earlier numerical study of the Kussoy-Horstman configuration. A comparison of the current results with those of Ref. 15 should give some insight regarding the required resolution for grid-independent results.

Representative Mach number contours for the Kussoy-Horstman geometry are shown in Figure 1 (97x65x89 grid, Edwards-McRae model). The fin-generated oblique shock separates the flat-plate viscous layer, producing a vortical structure that grows in a nearly conical fashion for a short distance away from the fin leading edge. Boundary layer displacement effects result in the formation of separation and rear shock surfaces, further complicating the picture. An expansion fan caused by the curving of the external flow around the thickened viscous layer is also observed in the region between the fin and the planar shock surface.

In Figure 2, crossflow surface pressure distributions corresponding to each of the models are compared with the experimental data at $X = 18.2$ cm. The lateral extent of the crossflow vortical structure and the magnitude of the separation-shock pressure rise are predicted well by all computations, while the peak pressure level near the fin ($Z = 0$) is consistently underpredicted. Minor differences among the model predictions are evidenced in Fig. 3, a plot of the normalized lateral heat transfer at $X = 16.45$ cm. The rise in heat transfer near the fin surface, a consequence of the movement of the high-temperature external

flow behind the shock to the surface via vortex action, is overpredicted by all approaches. A second peak in heat transfer at $Z \approx 6$ cm, a possible result of a secondary separation, is not captured by any of the computations. In general, the Spalart-Allmaras solution displays the best agreement with the measured heat transfer level near the fin, while the Edwards-McRae and Edwards-Chandra solutions more accurately capture the lateral extent of the heat transfer rise. The Edwards-McRae and Edwards-Chandra models provide essentially the same result, indicating that the effects of the nonequilibrium modifications are insignificant for situations in which the directions of the maximum pressure gradient and the maximum velocity component do not coincide. The flat plate viscous layer within the interaction region remains in local equilibrium, at least as defined by Eqns. 10-16.

Figure 4 compares the measured skin friction distribution at $X = 15.5$ cm with computed distributions obtained using the relation

$$C_f = \frac{\mu \sqrt{(\frac{\partial w}{\partial Y})^2 + (\frac{\partial u}{\partial Y})^2}|_{wall}}{\frac{1}{2} \rho_{\infty} U_{\infty}^2} \quad (29)$$

The solutions obtained using the Baldwin-Barth, Edwards-McRae, and Edwards-Chandra models agree quite well with the rather sparse experimental data set, particularly near the fin, while the Spalart-Allmaras result underpredicts the peak skin friction level. The consistency of this behavior with that evidenced in Figure 3 indicates that the computed flow in the corner region behaves in accordance with the "Reynolds analogy" created by the combination of the Boussinesq hypothesis and the constant turbulent Prandtl number assumption.

The surface pressure distribution along the fin at $X = 17.72$ cm is illustrated in Figure 5. Again, generally good agreement with the experimental data is evidenced for all of the computations. Figure 6 compares the computed heat transfer distributions along the fin surface with the experimental data measured at $X = 16.13$ cm. The experimental distribution indicates that the fin boundary layer is laminar at the measurement station, except for a region near the plate surface ($Y = 0.0$ cm). None of the eddy viscosity - transport approaches predict a purely laminar boundary layer on the fin. As shown in Ref. 15, however, this inaccuracy does not influence the surface properties along the plate to any large extent. Among the closure options, the Baldwin-Barth model appears to produce the most abrupt transition, while the Edwards-McRae variants and the Spalart-Allmaras model result in a more delayed response. As pointed out in Ref. 6, a lowering of the constant β in the damping function argument (8) can significantly affect the "numerical" transition distance predicted by the Edwards-McRae model. The value of this constant, 0.005, was chosen to ensure that the experimentally-determined inflow boundary layer properties were matched at the proper distance from the flat plate leading edge (187 cm). This value

was left unchanged for the actual interaction computation, leading to the results shown in Figure 6.

As shown in Figures 2-6, the effects of a two-fold increase in grid resolution on the solution obtained using the Edwards-McRae model are essentially negligible. Similar trends would probably hold for the other turbulence models. Figure 7 compares the convergence histories for the Edwards-McRae computation using both the $97 \times 89 \times 65$ and the $113 \times 97 \times 113$ grids. As indicated, the multigrid iteration strategy results in grid-independent convergence behavior to four decimal places in the residual norm, measured relative to its initial value. The convergence rates for the other turbulence models on the coarser grid (not shown) follow a very similar pattern. Higher levels of residual reduction could be achieved without difficulty; however, changes in the outputted surface quantities were negligible past the four order-of-magnitude reduction point.

Trends similar to those evidenced in Figures 1-6 were also observed by Knight, et al.¹⁵, who employed the Jones-Launder $k - \epsilon$ model and a two-layer $k - \epsilon / k - L$ model for their full Navier-Stokes simulations. Given the differences in the level of mesh resolution and in the turbulence models, the agreement between the computed results of Ref. 15 and the results of the present investigation is rather remarkable. In particular, the earlier simulations also overpredict the peak surface heat transfer level while predicting accurately the peak level of skin friction. Eliminating, for the time being, the possible need for an explicit "compressibility correction" in all models and the possibility of significant experimental error, it would appear that the major source of the disagreement between theory and experiment for this flow lies not in the choice of turbulence model nor in the grid resolution, but in the accuracy of the basic Boussinesq constitutive relation in the corner region.

6.2 Wideman, et al. cylinder-flare

The second database utilized in the evaluation of the eddy viscosity - transport closures corresponds to a recent experiment of Wideman, et al.¹² The basic geometry consists of a blended cylinder - 20° cone combination in which the centerline of the cone is displaced from the centerline of the initial cylinder section by a distance of 1.27 cm (half the radius of the initial cylinder section). An $X - R - \theta$ coordinate system is used to facilitate the presentation of the computational and experimental results. The top and bottom of the cylinder are associated with $\theta = 0^\circ$ and $\theta = 180^\circ$, respectively, while the origin of the axial coordinate X is located at the innermost $\theta = 0^\circ$ cylinder-cone juncture. Relative to this point, the axial displacement of the innermost juncture line is defined by

$$X = \frac{\hat{R} \sqrt{5 - 4 \cos \theta} - 1}{2 \tan 20^\circ} \quad (30)$$

where $\hat{R}$ is the radius of the initial cylinder section (2.54

cm). Another juncture line, displaced 10.47 cm downstream of the innermost juncture, defines the intersection of the cone with a larger cylinder (6.35 cm radius). A basic outline of the geometry is given in Figure 8; Ref. 12 should, however, be consulted for further details.

The free-stream conditions and the inflow boundary layer properties are summarized in the following table:

	Wideman, et al. cylinder-flare
Mach no.	2.89
Re/meter	1.7×10^7
T_∞ ($^\circ K$)	105.0
T_{wall}/T_∞	adiabatic
Inflow profile location (cm)	-2.0
Inflow δ (cm)	1.10
Inflow C_f	1.44×10^{-3}

The inflow boundary layer profiles for each computation were determined by solving the axisymmetric Navier-Stokes equations over the initial cylinder section until the conditions in the preceding table were matched to sufficient accuracy.

Two grids, one with 97x89x65 points in the X , R , and θ directions and one with 113x89x113 points, are utilized in this study. The minimum radial grid spacing for both grids is 1.37×10^{-6} meters, fine enough to ensure an inflow $\hat{n}_{min}^+$ of less than unity. Points are also clustered in the vicinity of the innermost cylinder-cone juncture to resolve elements of the flow topology. As a result of the emphasis on the initial interaction region, the expansion region aft of the outermost cylinder-cone juncture is slightly under-resolved in the X direction.

Mach number contours in the $\theta = 0^\circ$ and $\theta = 180^\circ$ symmetry planes and at three streamwise stations are shown in Figure 8 (97x89x65 grid, Edwards-McRae model). The dominant inviscid feature of the interaction is the oblique shock surface formed due to the abrupt change in the slope of the geometry aft of the innermost juncture line. Because of the asymmetry of the configuration, the shock compresses the flow first at the top ($\theta = 0^\circ$), inducing a pressure gradient in the circumferential direction. The nonuniform pressure distribution results in a general movement of the flow behind the shock surface from $\theta = 0^\circ$ to $\theta = 180^\circ$. Associated with the development of the velocity component in the circumferential direction is the formation of a vortical structure that is roughly aligned with the innermost cylinder-cone juncture. Due to the lateral symmetry of the flow at $\theta = 180^\circ$, the circumferential component of velocity is retarded, leading to a transfer of momentum from the circumferential component to the radial and axial components. The result is a thickening of the boundary layer on the bottom of the configuration and the formation of a region of axially-separated flow upstream of the cylinder-cone juncture.

Another abrupt change in the surface slope at the outermost cylinder-cone juncture induces the formation

of an expansion surface. At the outermost juncture, the boundary layer is primarily supersonic, and the effects of the expansion are not felt in the primary interaction region. This behavior has been confirmed by calculations on another grid system that only encompasses the initial cylinder section and the cone.

Computed surface pressure distributions at five azimuthal stations, $\theta = 0^\circ, 45^\circ, 90^\circ, 135^\circ$, and 180° , are compared with the measured data in Figure 9. The computed pressure distributions (as well as the skin friction distributions discussed next) correspond to the 97x89x65 grid. No significant differences in either the surface properties or the major topological features were observed for the calculation performed on the larger grid using the Edwards-McRae model. As before, it is likely that a similar level of grid-independence would hold for the other models as well. As shown in Figure 9, all computations accurately predict both the general shape of the pressure distribution in the compression region and the upstream extent of the separation-shock pressure rise. Near the rearmost cylinder-flare juncture, all solutions underpredict slightly the measured pressure level, an effect possibly the result of insufficient streamwise grid resolution. For all azimuthal stations, the pressure distributions provided by the Edwards-McRae, Edwards-Chandra, and Spalart-Allmaras models essentially collapse upon one another, providing another indication of the inherent similarity of the approaches in terms of predictive capability. The Baldwin-Barth solution underpredicts slightly the rate of pressure recovery downstream of the initial rise.

As in the Kussoy-Horstman configuration, more substantial differences among the model predictions occur with regard to the skin friction distributions (Figure 10). At $\theta = 0^\circ$, the solutions provided by the Edwards-McRae, Edwards-Chandra, and Spalart-Allmaras models accurately predict both the initial decrease in skin friction at the shock impingement location and the subsequent recovery of the skin friction downstream of the reattachment location. In contrast with 2-D results presented previously⁶, the rapid recovery of the near-wall velocity is due not only to an amplification in the turbulent fluctuation intensity caused by the shock wave but also to a general thinning of the boundary layer caused by the azimuthal movement of the flow. The Baldwin-Barth solution underpredicts significantly the rate of skin friction recovery downstream of reattachment, a trend very consistent with the results of Ref. 6.

At $\theta = 90^\circ$, significant near-wall velocity components in both the axial and circumferential directions are present. The flow does not separate in the sense of a reversed axial flow; rather, the effect of the nonuniform pressure distribution is to transfer momentum from the axial component to the circumferential component. All computations accurately predict the initial decrease in skin friction that delineates the forward extent of the crossflow vortical structure. However, all solutions dis-

play a more rapid recovery of the streamwise component of the skin friction than is observed experimentally. In contrast with the results at $\theta = 0^\circ$, the Baldwin-Barth computation provides the more accurate prediction of the rate of skin friction recovery. The discrepancy between the Baldwin-Barth prediction and that of the other models may be a result of the modeled form of the “destruction of eddy viscosity” term. For the Spalart-Allmaras model and the Edwards-McRae variants, the destruction term only depends on eddy viscosity variations in the radial direction, due to its explicit dependence on the distance from the surface. While this behavior is correct for a 2-D or axisymmetric configuration, it may be inadequate for flows exhibiting significant spanwise motion. The destruction term for the Baldwin-Barth model is invariant to coordinate rotation and thus depends on the eddy viscosity variation in both the radial and circumferential directions.

At $\theta = 180^\circ$, the 2-D nature of the interaction is evident, and the skin friction predictions behave in a manner consistent with the trends described in Ref. 6. In particular, all computations predict the location of the separation point accurately but produce a “stronger” separation (indicated by a larger magnitude of skin-friction coefficient) than is observed experimentally. The latter trend may contribute to the observed overprediction of the reattachment distance and the underprediction of the rate of recovery of the velocity field downstream of reattachment. The Spalart-Allmaras and Edwards-McRae distributions are very similar, while the Baldwin-Barth distribution deviates the most from the experimental data in the post-reattachment region. In accordance with the 2-D results of Ref. 6, the Edwards-Chandra model provides a modest improvement in the recovery of the skin friction but still fails to induce the correct trends within and just downstream of the separation region.

Figure 11 compares simulated surface oil flow patterns in the interaction region with the postulated topology of Ref. 12. The computed patterns were determined by mapping the axial and circumferential components of velocity at the first grid station above the surface to the $X = \theta$ plane. Several singular (or critical) points, locations at which the surface shear stress vanishes and the local flow direction is non-unique, are clearly indicated in all of the computed patterns, as are lines of coalescence which demarcate the extent of the separation region. As shown in Ref. 16, among others, it is possible to determine directly the character of the singular points by applying a phase-space analysis to the surrounding flow-field. We postpone such a rigorous treatment until a later date and instead, rely on the more phenomenological classification analysis of Chapman¹⁷ to determine the nature of the computed singular points and their effect on the structure of the external flow.

One of the dominant features of the surface topology is a line of separation [LS] emanating from a saddle of separation at the top of the surface ($\theta = 0^\circ$) and culmi-

nating at a node of separation [NS] located at $\theta \approx 160^\circ$. For all models, good agreement with the experiment is observed in the positioning of the line of separation. The experimentally-determined structure at $\theta \approx 160^\circ$ consists of two nodes of separation joined by a saddle of separation. As mentioned in Ref. 17, such a construct will behave as a node of separation. Given their basic simplicity, it is not surprising that the eddy viscosity - transport approaches coalesce the node-saddle-node system into a single node. The refined $113 \times 89 \times 113$ grid mentioned earlier was specifically designed to resolve the region surrounding the node in hopes of capturing the node-saddle-node combination. The calculation using the Edwards-McRae model on the refined grid also resulted in a single node of separation, however.

The line of separation marks the upstream extent of the vortical structure mentioned earlier; a line of attachment [LA] emanating from a node of attachment [NA] at $\theta = 0^\circ$ defines the downstream extent of the structure. From $\theta = 0^\circ$ to $\theta \approx 100^\circ$, good agreement with the experiment in the positioning of the line of attachment is again observed for all computations. At $\theta \approx 100^\circ$, the reattachment line splits into two lines of coalescence, one culminating in the node of separation and one continuing toward the $\theta = 180^\circ$ centerline. This bifurcation, characteristic of all of the closure options but not indicated in the experimentally-determined topology (Figure 11e), represents the start of the breakdown of the tightly-organized crossflow vortical structure. At this juncture, part of the near-wall flow moves toward the node, where it detaches from the surface. The remainder continues toward the symmetry plane, where it recirculates before moving downstream.

Near the $\theta = 180^\circ$ reattachment point, two distinct topological structures are present in the computed surface patterns, neither of which is completely consistent with the postulated experimental structure. For the Spalart-Allmaras and Edwards-McRae solutions, the rearmost line of attachment terminates into a saddle point located at the $\theta = 180^\circ$ centerline (see Figures 11a, 11c, and 12a). This saddle is connected to the saddle of separation that marks the upstream extent of the axial separation bubble. At the rearmost saddle, the near-wall flow bifurcates, some moving directly downstream and some moving upstream, either to depart from the surface at the forward saddle point or to recirculate before exiting through the node of separation at $\theta \approx 160^\circ$. Another line of coalescence, nearly parallel to the lower centerline, marks this division in the axially-separated flow (Figures 11a and 11c). As shown in Figure 12c for the Edwards-McRae solution, the rearmost saddle point is “nodelike” in the $\theta = 180^\circ$ plane of symmetry, and the flow is away from the surface. According to Chapman¹⁷, such a topology implies that the rearmost critical point is a saddle of separation. Thus, on the lower surface, the Edwards-McRae and Spalart-Allmaras models both produce a saddle-of-separation to saddle-of-separation connection, a topol-

ogy that is thought to be unstable except possibly under conditions of strong symmetry.¹⁷ The flow pattern in the lower symmetry plane (Figure 12c) indicates that much of the entrained near-wall flow departs from the rearmost saddle and moves upstream before recirculating and moving downstream. The separation is open-ended, as opposed to the closed-loop separation characteristic of a purely axisymmetric flowfield.

The surface topology predicted by the Baldwin-Barth and Edwards-Chandra computations near the $\theta = 180^\circ$ reattachment point is an example of a "pitchfork" bifurcation¹⁷ caused, evidently, by differences in the eddy viscosity fields. Here, the saddle of separation predicted by the other simulations is replaced by a node of attachment and two new saddles of separation located on either side of the centerline. Due to symmetry, only one of the saddles is shown in Figures 11b, 11d, and 12b. Associated with the differences in type (attachment vs. separation) among the critical points is an additional critical point located off the surface within the centerline plane (see Figure 12d). While the separation in the lower symmetry plane is obviously still open-ended, the details of the near-wall flow structure are considerably more complex than for the Edwards-McRae / Spalart-Allmaras topology. In particular, it appears that some of the circumferentially-moving fluid departs from the surface and enters the embedded critical point. From this location, some of the entering fluid moves upstream within the symmetry plane before recirculating, while some moves directly downstream. The node of attachment on the surface is "fed" by one of the rearward-moving streamlines, leading to another complex recirculation pattern in the vicinity of the node.

The authors of Ref. 12 maintain that the rearmost line of attachment culminates in a saddle point. However, they conclude that the rearmost saddle is a saddle of attachment, a topology that would imply the existence of a closed-loop separation bubble in the $\theta = 180^\circ$ symmetry plane. Such a connection would be extremely sensitive to small perturbations, yet the authors state that the saddle-to-saddle connection passed numerous tests of repeatability.¹⁸ They also state, however, that their oil flow resolution might not have been high enough to capture other critical points in the reattachment region and that their "saddle of attachment" conclusion was based heavily on the surface pattern.^{12,18} Based on these comments and given the obvious weaknesses of all of the models in the prediction of the skin friction response in the reattachment / recovery region, it is not clear at this point which (if any) of the postulated topologies most closely represents the actual flowfield. If the saddle-saddle connection is accepted, however, the topology produced by the Edwards-McRae and Spalart-Allmaras models does provide a plausible alternative to the structure detailed in Ref. 12. Comparisons of the eddy viscosity - transport results with those of other turbulence closure options might shed further light on the situation, but only the availabil-

ity of a more detailed experimental data set that includes velocity measurements in the lower symmetry plane will resolve the dilemma entirely.

Convergence histories corresponding to the Edwards-McRae simulations of the Wideman, et al. flow are illustrated in Figure 13. As before, the upwind relaxation multigrid algorithm performs well, reducing the residual error on the coarser mesh by almost five orders-of-magnitude in 100 cycles (approximately 1.6 hours of Cray Y-MP CPU time). On the finer $113 \times 89 \times 113$ mesh, a similar level of reduction requires 150 cycles (approximately 4.6 CPU hours), indicating that grid-independent convergence rates are not quite achieved for this particular sequence of meshes. The pronounced "hump" in the convergence distributions at approximately three orders-of-magnitude reduction probably indicates the start of the growth of the axial separation bubble on the lower surface. Similar convergence distributions for 2-D axially-separated flows have been noted previously.⁶

7. Conclusions

This work completes an effort designed to test the performance of four one-equation "eddy viscosity - transport" turbulence models in computing strongly-interacting compressible flows in two and three dimensions. Two recent experiments, one involving the Mach 8 interaction of a fin-generated oblique shock with a developing flat-plate boundary layer and the other involving Mach 3 flow over a cylinder-flare configuration, have been numerically simulated using a recently-developed multigrid approach. Fine grids have been utilized throughout to reduce uncertainties associated with the discretization error.

The current results indicate that all of the tested one-equation models (Baldwin-Barth, Edwards-McRae, Spalart-Allmaras, and Edwards-Chandra) provide good agreement with the measured pressure distributions, particularly in the prediction of the upstream (or lateral) extent of the separated boundary layer. In concert with previous two-dimensional results, more significant deviations from the experimental data and from each other occur for the skin friction and heat transfer distributions. For the sharp fin interaction, all of the tested closures overpredict the peak heat transfer level near the fin - flat plate juncture, a trend very consistent with the results of other numerical simulations using different closure approaches and different levels of grid resolution. For this flow, the direction of the maximum pressure gradient and the direction of the dominant velocity component do not coincide, and, as a result, no significant differences between the Edwards-Chandra and Edwards-McRae predictions are observed.

For the cylinder-flare configuration, the success of the models in predicting the skin friction distribution is rather variable. While the Baldwin-Barth model induces a very poor rate of skin friction recovery in regions where the "two-dimensional" nature of the interaction is

predominant, it performs better than the other models in regions where both axial and circumferential components of the near-wall velocity are important. In regions where the flow is axially-separated, the model predictions are in accord with trends evidenced earlier in two-dimensional simulations. In particular, all models produce a "stronger" separation than is observed experimentally, an effect possibly due to the near-wall calibration in terms of the log-law relationship. An associated effect is a rather poor prediction of the recovery rate of the near-wall flow downstream of the reattachment point. The Baldwin-Barth model fares worst in this regard, while the Edwards-Chandra approach performs best.

A preliminary analysis of the structure of the cylinder - flare flow has also been performed. Most of the topological features present in the experimental surface oil flow patterns are also found in the computed patterns, and good agreement in the positioning of the features is generally observed. However, differences in interpretation make further study necessary before a complete understanding of the flow structure can be achieved. It is of some interest that the Edwards-McRae and Spalart-Allmaras computations both predict a rare saddle-to-saddle connection on the bottom of the configuration, a feature that is in accord with the experimental results. The other models result in a "pitchfork" bifurcation of the rearmost saddle point into a node, two saddles offset from the centerline, and a critical point located above the surface.

It is apparent from the current results and from the results of earlier efforts that the tested eddy viscosity - transport closures offer a good level of accuracy and predictive capability for strongly compressible, shock-separated flows without incurring a great expense. No model has shown itself to be markedly superior to the others - all have similar weaknesses and similar strengths. For shock wave / boundary layer interactions, the most prevalent weakness in the eddy viscosity - transport closure hypothesis is a tendency to induce an overly-rapid initial response to a strong disturbance in the inner part of a turbulent boundary layer. The result, further downstream, is a slow rate of recovery from the effects of the disturbance. Alternative calibrations for the near-wall region, based on something other than the logarithmic law in adverse pressure gradient situations, may have to be considered in order to alleviate this difficulty.

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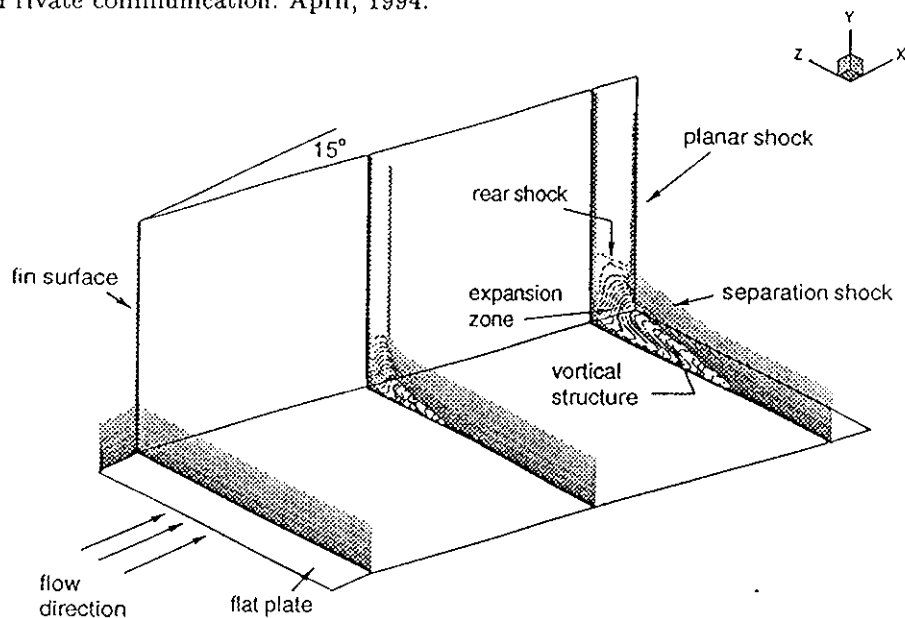


Fig. 1: Mach number contours (Kusoy 15° fin)

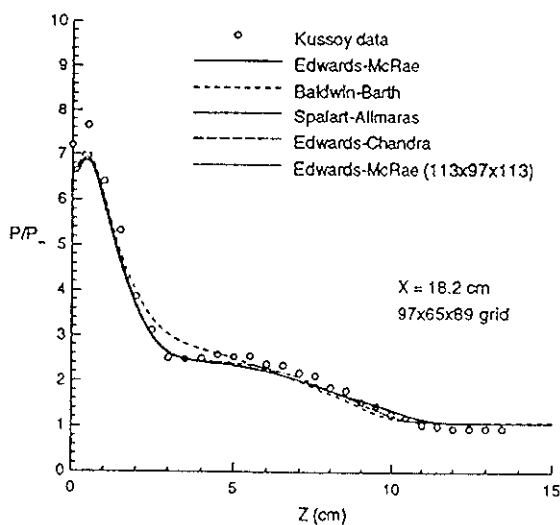


Fig. 2: Plate pressure (Kusoy 15° fin)

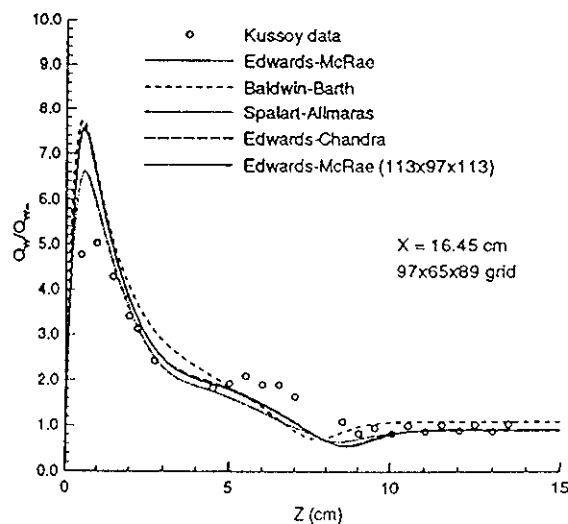


Fig. 3: Plate heat transfer (Kusoy 15° fin)

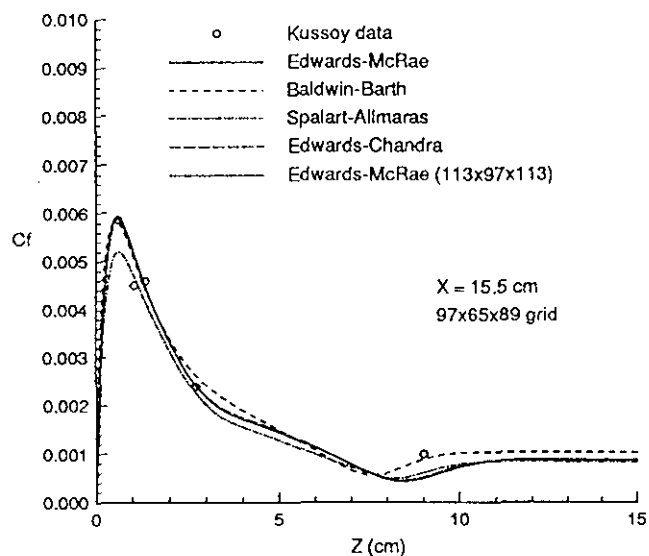


Fig. 4: Plate skin friction (Kussoy 15° fin)

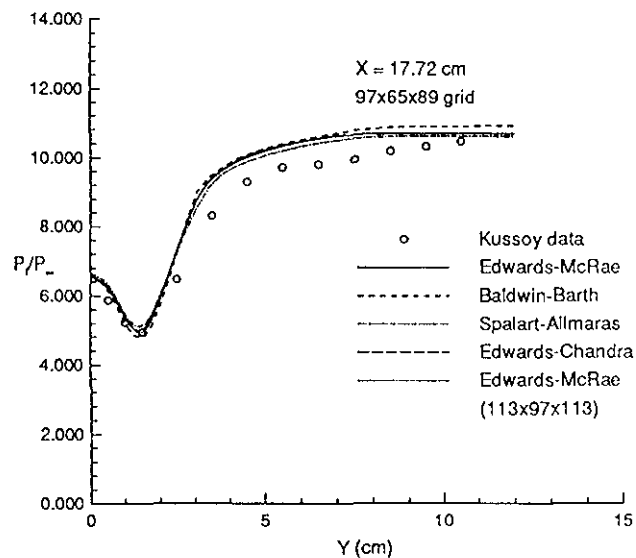


Fig. 5: Fin pressure (Kussoy 15° fin)

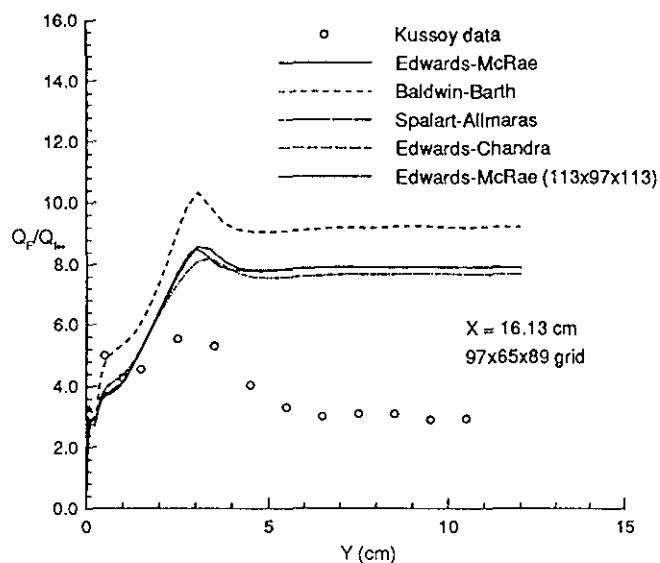


Fig. 6: Fin heat transfer (Kussoy 15° fin)

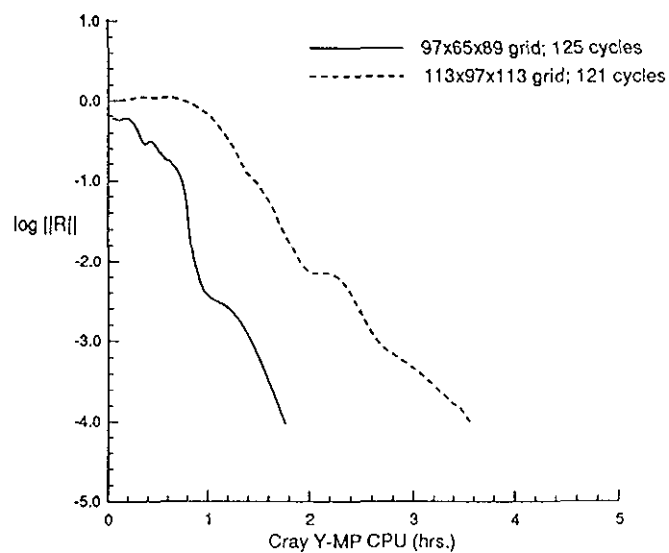


Fig. 7: Convergence histories (Kussoy 15° fin)

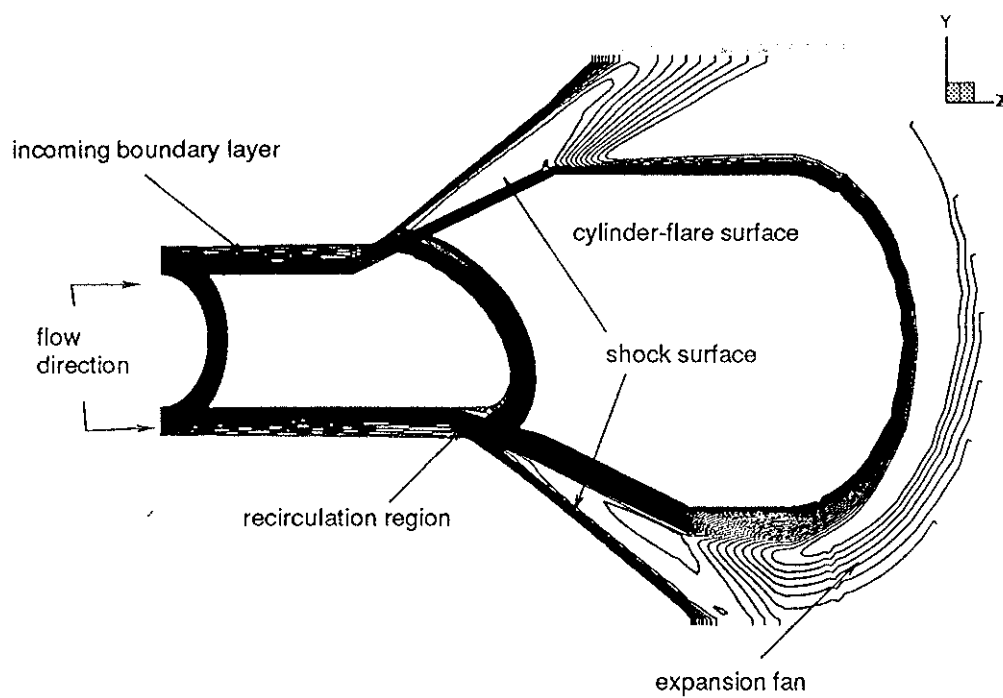


Fig. 8: Mach number contours (Wideman 20° flare)

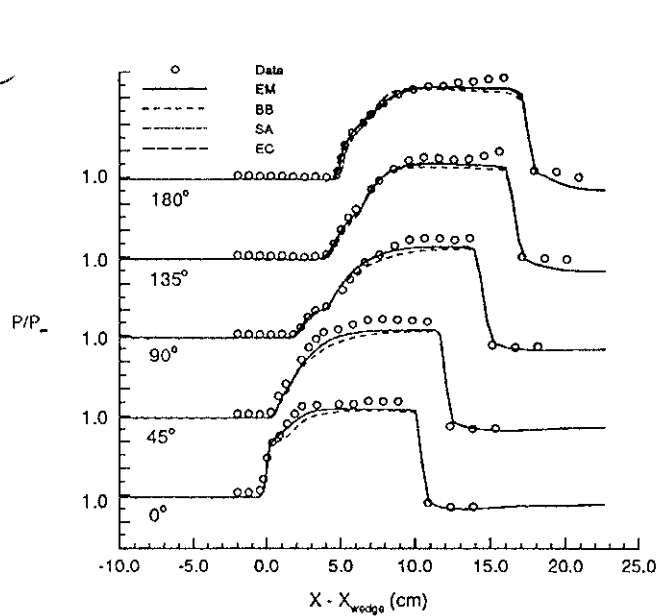


Fig. 9: Surface pressure (Wideman 20° flare)

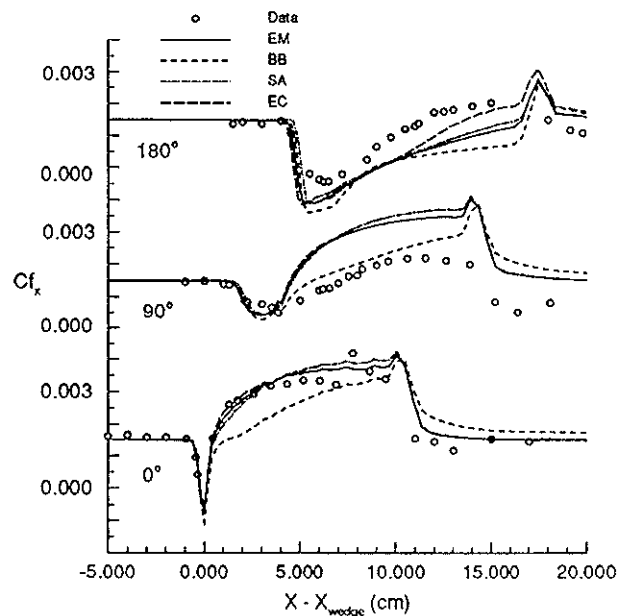
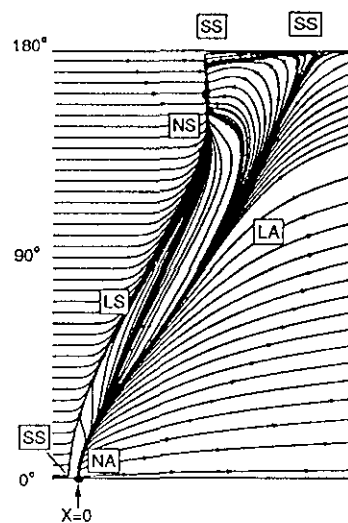
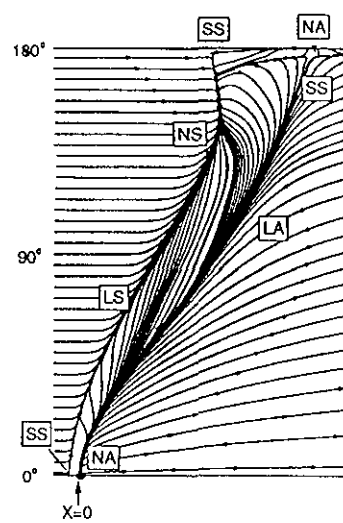


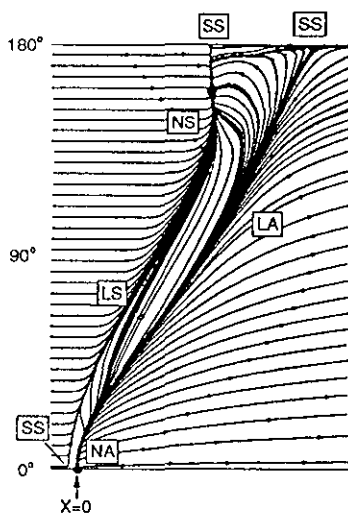
Fig. 10: Skin friction (Wideman 20° flare)



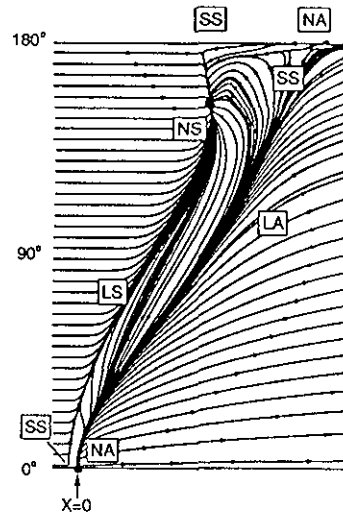
a: Edwards-McRae



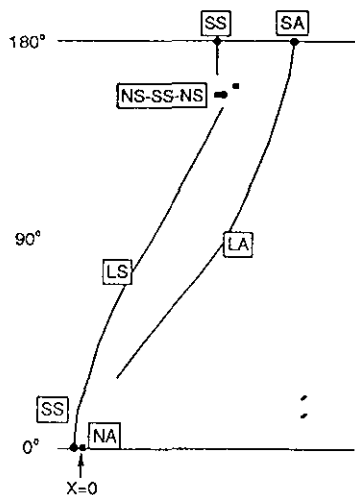
b: Baldwin-Barth



c: Spalart-Allmaras



d: Edwards-Chandra

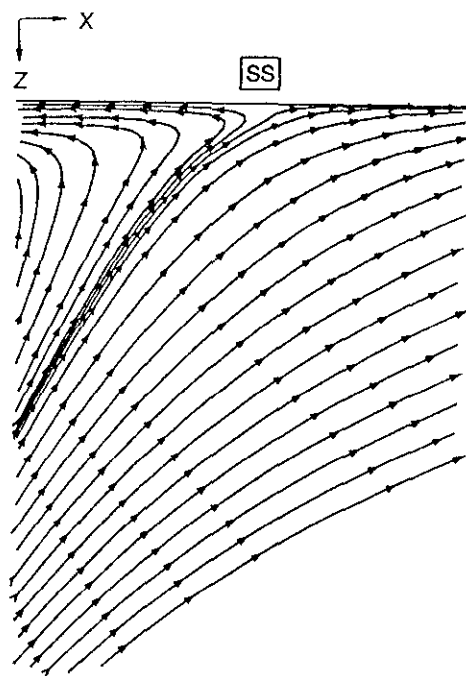


e: experiment

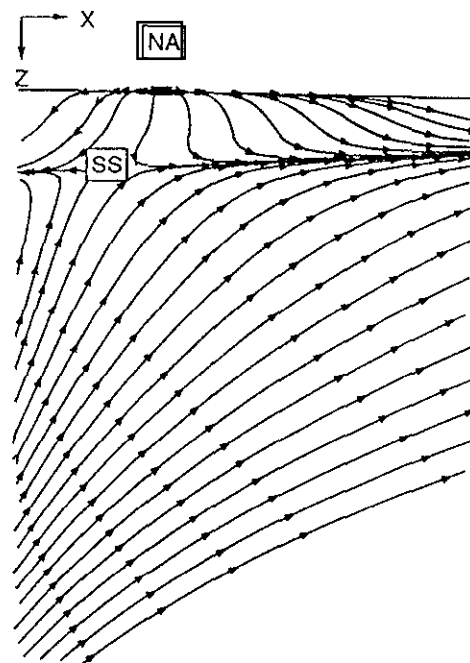
Nomenclature:

- LA -- line of attachment
- LS -- line of separation
- NA -- node of attachment
- NS -- node of separation
- SS -- saddle of separation
- SA -- saddle of attachment
- CP -- critical point

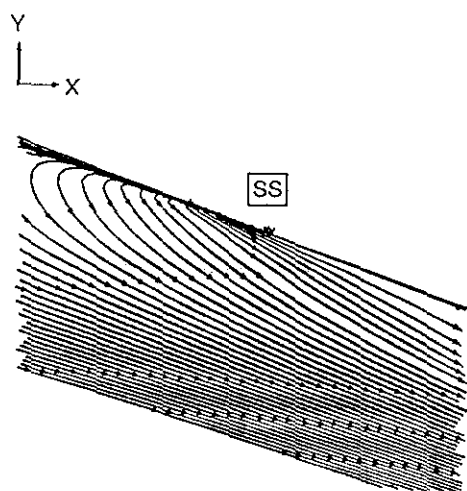
Fig. 11: Surface oil flow patterns (Wideman 20° flare)



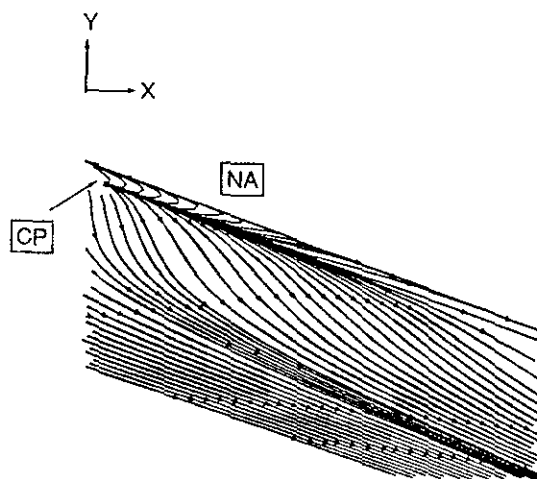
a. Edwards-McRae (X-Z)



b. Baldwin-Barth (X-Z)



c. Edwards-McRae (X-Y)



d. Baldwin-Barth (X-Y)

Fig. 12: Closeup views: streamline patterns near reattachment ($\theta = 180^\circ$)

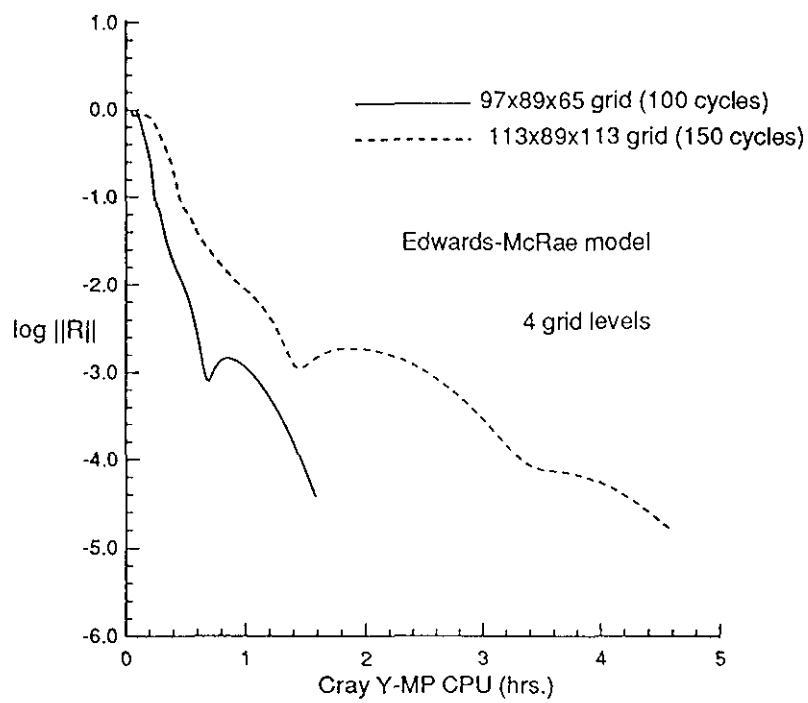


Fig. 13: Convergence histories (Wideman 20° flare)